



Tecnológico Nacional de México

Instituto Tecnológico de Saltillo

Computo en la nube

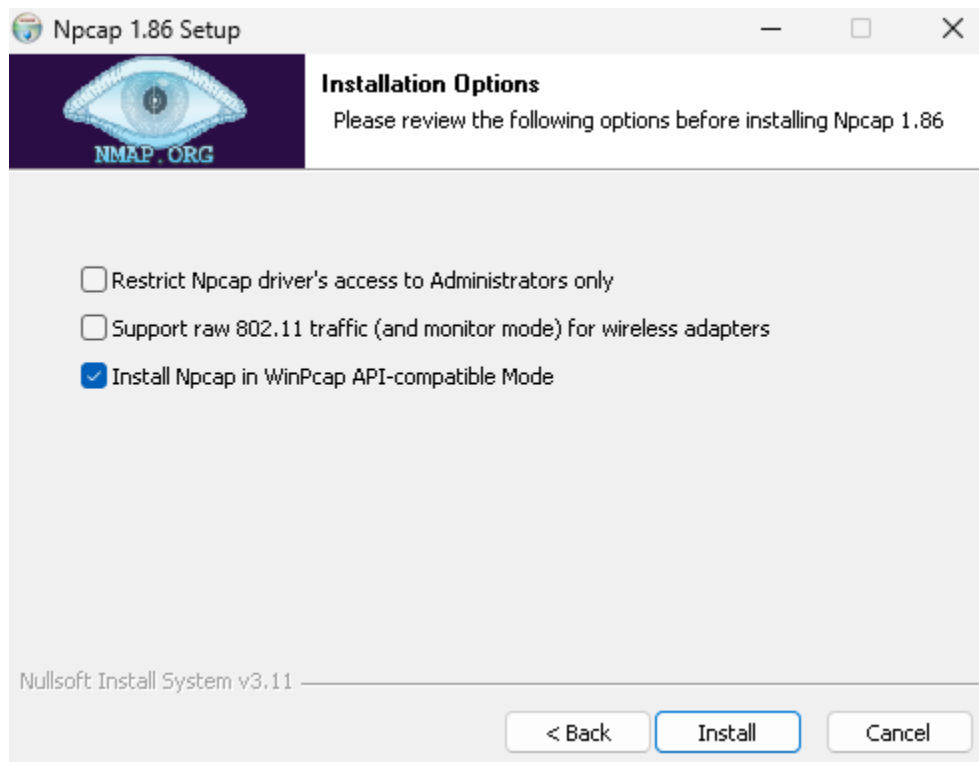
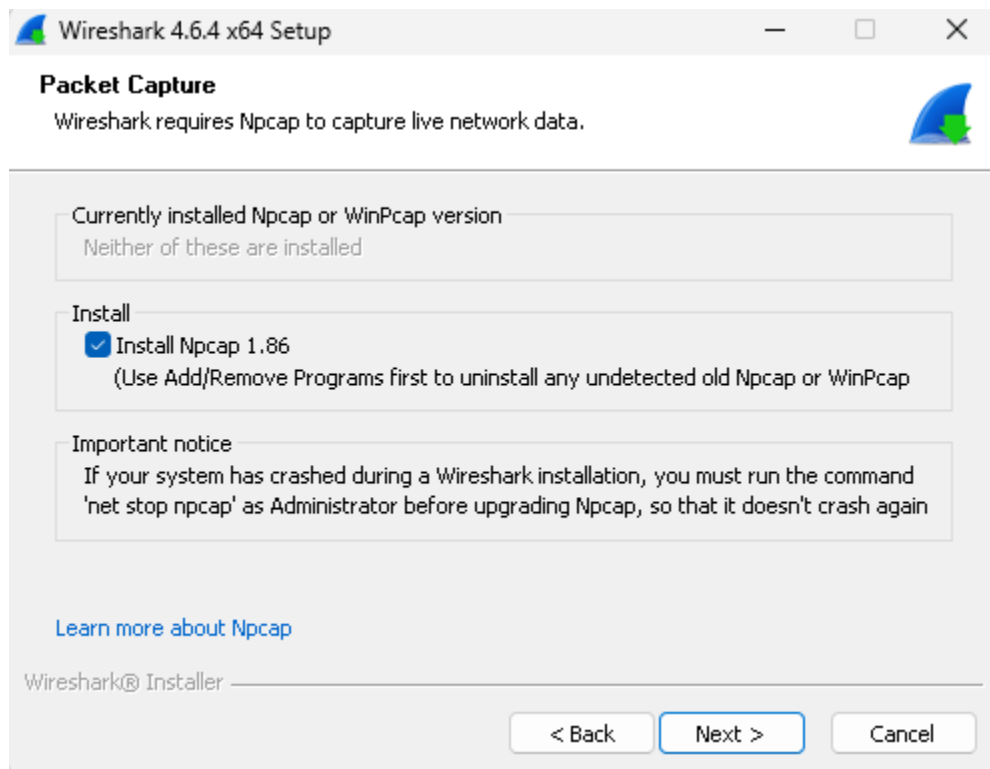
4.10 Análisis de trafico de red con Wireshark para la prevención de ataques

Docente: Miguel Salazar Del Bosque

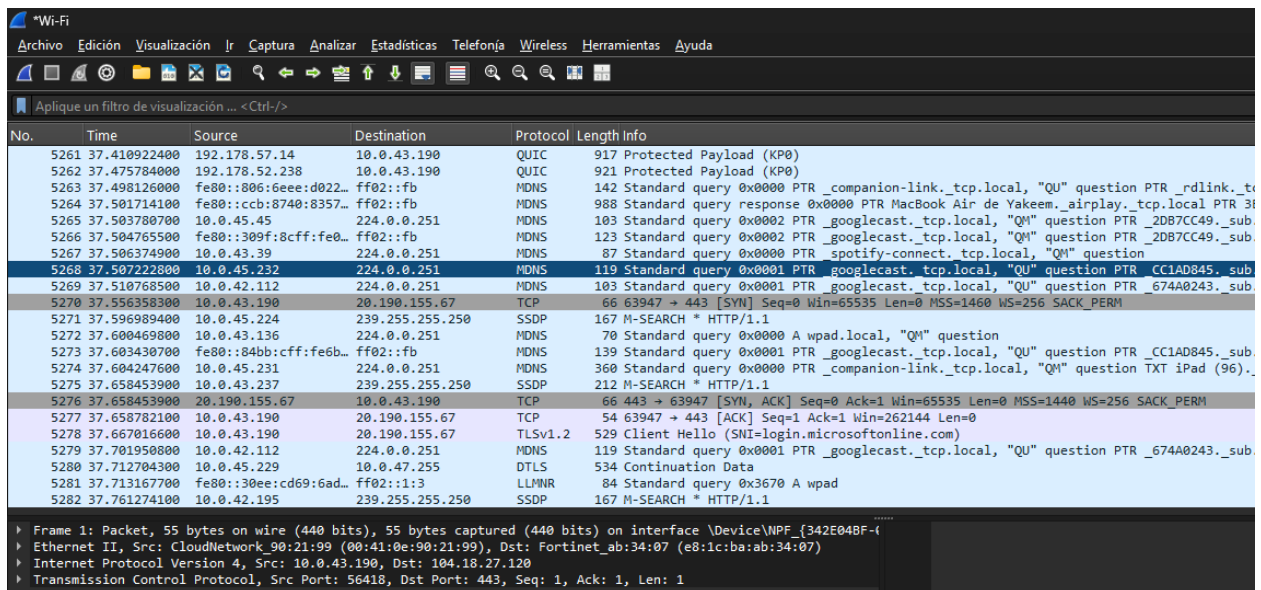
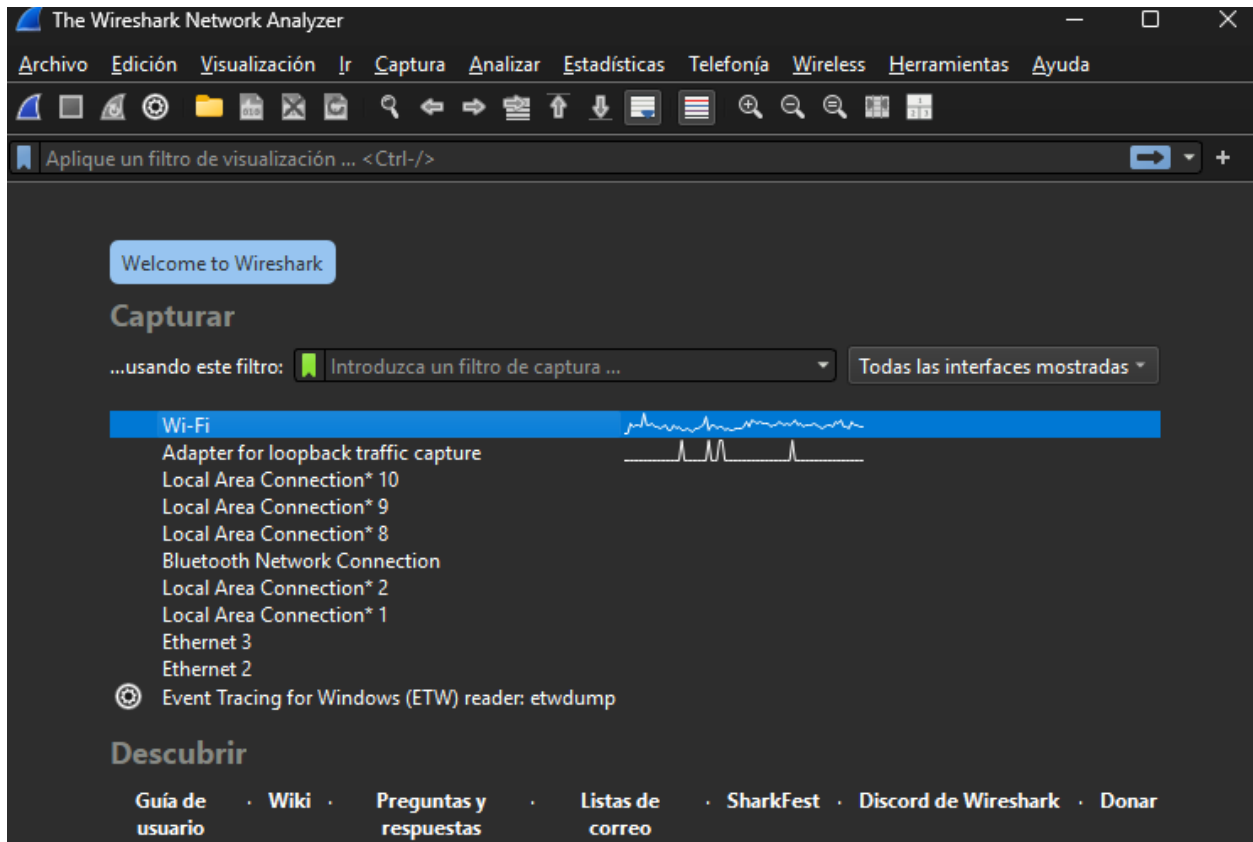
**Alumno
Diego Josué González de León**

25 de marzo de 2026

Instalar Wireshark con Npcap

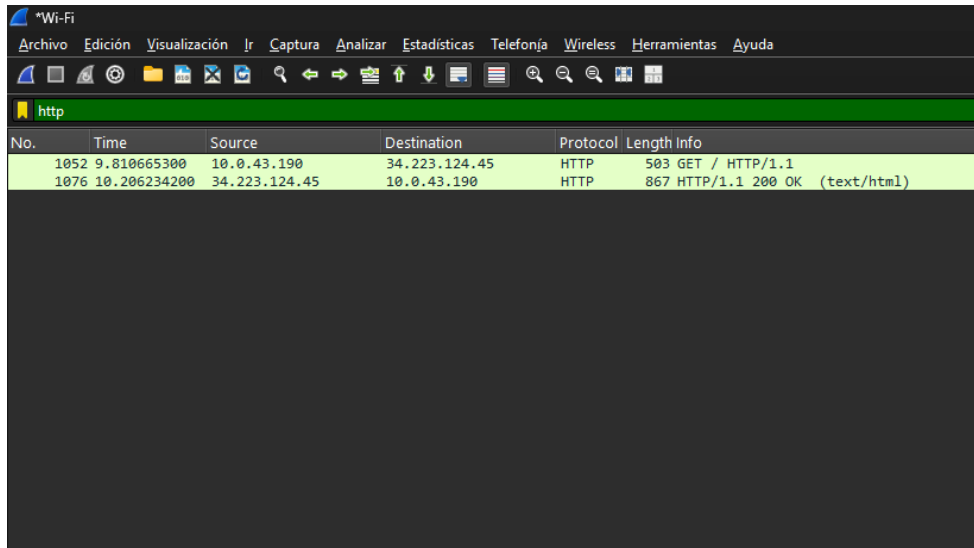


Abrir Wireshark en la interfaz Wi-Fi y empezar a capturar trafico



Verificar el tráfico filtrado

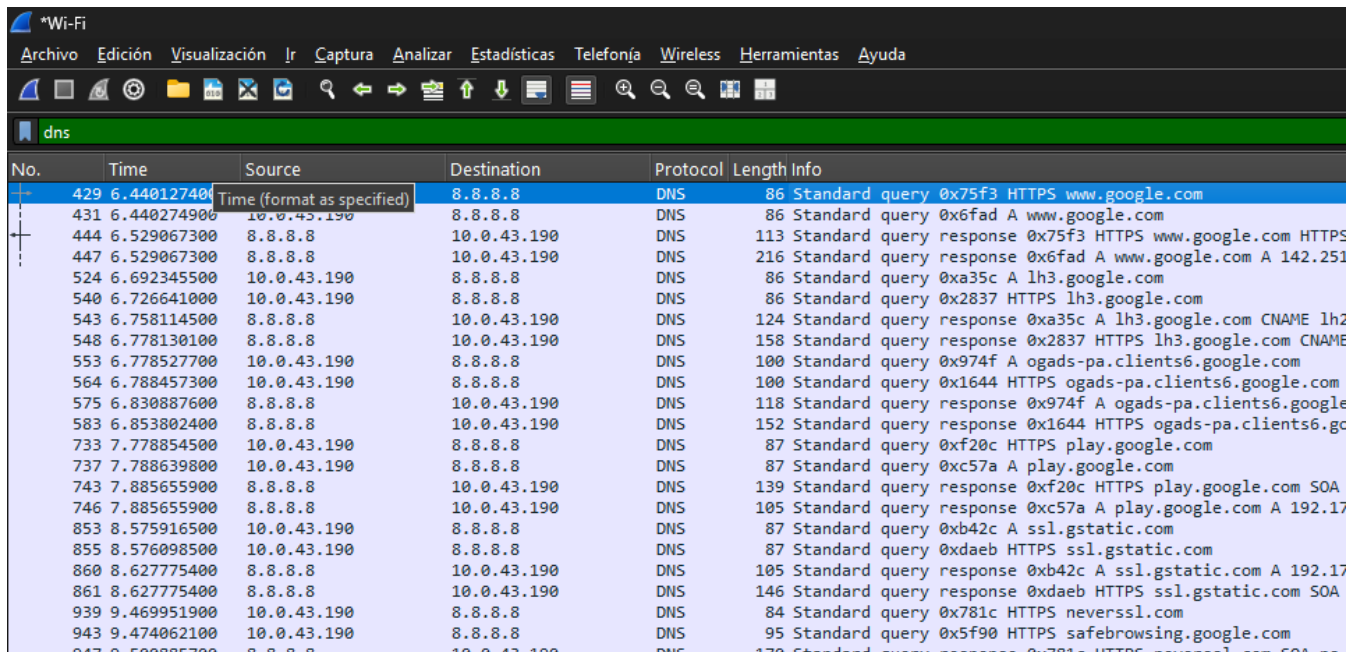
HTTP



The image shows a network monitoring application window titled '*Wi-Fi'. The menu bar includes 'Archivo', 'Edición', 'Visualización', 'Ir', 'Captura', 'Analizar', 'Estadísticas', 'Telefonía', 'Wireless', 'Herramientas', and 'Ayuda'. The toolbar contains various icons for file operations, navigation, and analysis. The main display area is titled 'http' and shows a table of captured HTTP traffic.

No.	Time	Source	Destination	Protocol	Length	Info
1052	9.810665300	10.0.43.190	34.223.124.45	HTTP	503	GET / HTTP/1.1
1076	10.206234200	34.223.124.45	10.0.43.190	HTTP	867	HTTP/1.1 200 OK (text/html)

DNS



The image shows the same network monitoring application window, but now displaying DNS traffic. The menu bar and toolbar are identical to the HTTP view. The main display area is titled 'dns' and shows a table of captured DNS traffic.

No.	Time	Source	Destination	Protocol	Length	Info
429	6.440127400	10.0.43.190	8.8.8.8	DNS	86	Standard query 0x75f3 HTTPS www.google.com
431	6.440274900	10.0.43.190	8.8.8.8	DNS	86	Standard query 0x6fad A www.google.com
444	6.529067300	8.8.8.8	10.0.43.190	DNS	113	Standard query response 0x75f3 HTTPS www.google.com HTTPS
447	6.529067300	8.8.8.8	10.0.43.190	DNS	216	Standard query response 0x6fad A www.google.com A 142.251
524	6.692345500	10.0.43.190	8.8.8.8	DNS	86	Standard query 0xa35c A lh3.google.com
540	6.726641000	10.0.43.190	8.8.8.8	DNS	86	Standard query 0x2837 HTTPS lh3.google.com
543	6.758114500	8.8.8.8	10.0.43.190	DNS	124	Standard query response 0xa35c A lh3.google.com CNAME lh2
548	6.778130100	8.8.8.8	10.0.43.190	DNS	158	Standard query response 0x2837 HTTPS lh3.google.com CNAME
553	6.778527700	10.0.43.190	8.8.8.8	DNS	100	Standard query 0x974f A ogads-pa.clients6.google.com
564	6.788457300	10.0.43.190	8.8.8.8	DNS	100	Standard query 0x1644 HTTPS ogads-pa.clients6.google.com
575	6.830887600	8.8.8.8	10.0.43.190	DNS	118	Standard query response 0x974f A ogads-pa.clients6.google
583	6.853802400	8.8.8.8	10.0.43.190	DNS	152	Standard query response 0x1644 HTTPS ogads-pa.clients6.gc
733	7.778854500	10.0.43.190	8.8.8.8	DNS	87	Standard query 0xf20c HTTPS play.google.com
737	7.788639800	10.0.43.190	8.8.8.8	DNS	87	Standard query 0xc57a A play.google.com
743	7.885655900	8.8.8.8	10.0.43.190	DNS	139	Standard query response 0xf20c HTTPS play.google.com SOA
746	7.885655900	8.8.8.8	10.0.43.190	DNS	105	Standard query response 0xc57a A play.google.com A 192.17
853	8.575916500	10.0.43.190	8.8.8.8	DNS	87	Standard query 0xb42c A ssl.gstatic.com
855	8.576098500	10.0.43.190	8.8.8.8	DNS	87	Standard query 0xdaeb HTTPS ssl.gstatic.com
860	8.627775400	8.8.8.8	10.0.43.190	DNS	105	Standard query response 0xb42c A ssl.gstatic.com A 192.17
861	8.627775400	8.8.8.8	10.0.43.190	DNS	146	Standard query response 0xdaeb HTTPS ssl.gstatic.com SOA
939	9.469951900	10.0.43.190	8.8.8.8	DNS	84	Standard query 0x781c HTTPS neverssl.com
943	9.474062100	10.0.43.190	8.8.8.8	DNS	95	Standard query 0x5f90 HTTPS safebrowsing.google.com

TCP

No.	Time	Source	Destination	Protocol	Length	Info
406	6.323848200	10.0.43.190	13.223.19.204	TCP	54	TCP ACKed
410	6.323997100	172.64.148.235	10.0.43.190	TLSv1.2	133	Application
411	6.344705300	10.0.43.190	8.8.8.8	TCP	66	59260 → 53
412	6.345200100	10.0.43.190	8.8.8.8	TCP	66	65368 → 53
413	6.375011700	10.0.43.190	172.64.148.235	TCP	54	62655 → 443
422	6.439659100	8.8.8.8	10.0.43.190	TCP	66	53 → 59260
423	6.439659100	172.64.148.235	10.0.43.190	TCP	60	443 → 62655
424	6.439895500	10.0.43.190	8.8.8.8	TCP	54	59260 → 53
425	6.439983600	8.8.8.8	10.0.43.190	TCP	66	53 → 65368
427	6.440059600	10.0.43.190	8.8.8.8	TCP	54	65368 → 53
428	6.440063100	10.0.43.190	8.8.8.8	TCP	56	59260 → 53
429	6.440127400	10.0.43.190	8.8.8.8	DNS	86	Standard que
430	6.440230900	10.0.43.190	8.8.8.8	TCP	56	65368 → 53
431	6.440274900	10.0.43.190	8.8.8.8	DNS	86	Standard que
433	6.471203600	172.64.148.235	10.0.43.190	TLSv1.2	133	Application
434	6.471203600	8.8.8.8	10.0.43.190	TCP	60	53 → 59260
442	6.515675600	10.0.43.190	172.64.148.235	TCP	54	62655 → 443
443	6.529067300	8.8.8.8	10.0.43.190	TCP	60	53 → 59260
444	6.529067300	8.8.8.8	10.0.43.190	DNS	113	Standard que
445	6.529067300	8.8.8.8	10.0.43.190	TCP	60	53 → 65368
446	6.529067300	8.8.8.8	10.0.43.190	TCP	60	53 → 65368

Texto Claro

```

Wireshark - Seguir secuencia TCP (tcp.stream eq 23) - Wi-Fi

GET / HTTP/1.1
Host: neverssl.com
Connection: keep-alive
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Accept-Encoding: gzip, deflate
Accept-Language: es-US;es;q=0.9,en;q=0.8,en;q=0.7

HTTP/1.1 200 OK
Date: Wed, 25 Mar 2026 13:52:54 GMT
Server: Apache/2.4.66 ()
Upgrade: h2,h2c
Connection: Upgrade, Keep-Alive
Last-Modified: Wed, 29 Jun 2022 00:23:33 GMT
ETag: "f79-5e28b29d38e93-gzip"
Accept-Ranges: bytes
Vary: Accept-Encoding
Content-Encoding: gzip
Content-Length: 1906
Keep-Alive: timeout=5, max=100
Content-Type: text/html; charset=UTF-8

.....Mkn.....B#.....C.Z.k.R...q-...A...&.....Q...l...+...yr...l...u.PC%<...2H)UZ...7.N.....{...V...h3.12..1).2Z...C.Vk{...#...k.t#
*...+...0.R...{.T.&$...zB^?...9Pz.E^...x[m...=...Q%V...+...T.2\....xf/e5'1
...y.s...#.....?...r...V.e%...{U...^v...
...q...kQ...T...
...C...y/.C...l...u.1.....}...
...#k1>O...3.E...t...
...A.y...l...DP...h.K...2e...7;n...n.....V...>...R...b...LQ...m...p;$.>...e...1.B.....\...^Eq.x.2-.../D.....X
W...b.N...0...q1...G...1M...x...Kom(...N.W...f...Z7?...V...1.Z[...P...8a..3.1.T...Q.%:"Q...&...f.U...
K...1.&X...f...1...W/r/r/ET.....Io.h...*...#...h.1...>...2...b...g...0.R... \h{...d}E...z...u...
XR...")...7.S...>...0.r1...y)2^..
..E=B.O...j...o\...5$...
..6E..R.T..G+.....N.g3hZ.%)...6H.t...u...y.3.....gh;N7B...V.g?..e^..N...g..U.t<K.K.<
...j...d...N.1...6y...P+...g...1.....&...@b
7...=(...^r.....,=..M.=J...>@2<?...tu.....3.N...UdD.F^;Ohek1..b
Ge...c0..K9]7>...{.v.o.....&.l...u..m!o.v.g...{.j...g...X...
H...r...d...@...9m'...9...2B.67.....7.d...f.Zwiz...?..$d.X.....h...S)...q.M.onk...
(0...d...@...N...7.n...3].fb2...eQw...*#...C)P...P)...4.W...{q.h.....8Mk...e.....?.....h...x8
...S...{...2...X...5.....s...s...f)X.D.^...M.W;U.B...W...
N...y%...>...nM...a...>[d...P...sQc...?..{...:.....;E...=...4.fb..o78E[8.;G...T.C.)o]X.n'$.b!...+EF@...6.Fh...78.u.....S^...{...e.[C...c.W.
7Y9...%Reau...I...{.I1...n...h3)...tJo.p e..H...J.ID...@..t.1-#xm...t-Kq.l.1.=k..2...MX...z...%.G.Wb'...z...%.3...e9^"...@C.
C...{...p...f...f...{...1.y...l'...e.#1.....>...&x..H.6z..3Vk.a.8..oj.x0...o80...C...o0i0.O...N...1.....*3.r0...z...
.....H7...

Paquete 1052 (1906 bytes) en 1 turno(s). Clic para seleccionar.
Conversación completa (2722 bytes) Mostrar como ASCII No delta times Secuencia 23

```

¿Qué se ve en HTTP y no en HTTPS?

En HTTP puedes ver la información que se está enviando, por ejemplo contraseñas con sus usuarios, urls completas y diferentes parámetros.

En HTTPS no se puede visualizar el contenido cifrado. Solo se puede ver la IP y el dominio, pero no datos internos de la propia página.

¿Qué riesgos existen?

Cuando se accede a una página con http se puede filtrar nuestra información, ya que alguien que actúe con malicia podría interceptar la información de dicha página que nosotros estemos ingresando y esto significa que podrían acceder a contraseñas, datos personales, tarjetas de crédito/débito.

Es importante siempre verificar tanto la seguridad de la red a la que nos conectamos, es decir, quienes tienen acceso a ella, como también la seguridad intrínseca de los sitios web que visitamos, ya que nunca se sabe quién puede estar accediendo a nuestra información.